

Problem 8

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**Problem 1** (a) Let  $f$  be a positive differentiable function, defined on an open interval  $I$ . Find the formula for the derivative of the function  $\ln(f(x))$ .

(b) Using the formula obtained in part (a), compute the derivatives of the following functions.

(i)  $f(x) = \ln(x^2 + x + 1)$

(ii)  $f(x) = \ln(\sec(x) + \tan(x))$

(iii)  $f(x) = \ln(\ln(x))$